

Felix Dülmer

PhD Candidate in Computer Science · Technical University of Munich

 felixduelmer.github.io  felix-duelmer@online.de  github.com/Felixduelmer  Google Scholar

PROFILE

My research focuses on advancing medical and robotic imaging by incorporating physics into accurate, fast, and differentiable forward models used for simulation, inverse problems, and robotic acquisition. I build these with ray and path tracing for ultrasound, published at SIGGRAPH Asia, MICCAI, ICRA, MIDL, and IUS. PhD candidate at TUM (Prof. Nassir Navab) and visiting researcher at Stanford (Prof. Jeremy Dahl).

EXPERIENCE

PhD Candidate in Computer Science

01/2023 – Present

Technical University of Munich

- **Differentiable rendering:** built physics-based ultrasound renderers, from full-path ray tracing and scattering [7, 8] to differentiable path tracing for shape optimization [1] and refraction-aware speed-of-sound imaging [2].
- **Neural scene representations:** developed Gaussian ray casting for novel-view synthesis [4], multivariate Gaussian NeRFs for wide field-of-view reconstruction [3], occupancy networks for 3D reconstruction [9], and implicit neural deconvolution [10].
- **Robotic ultrasound:** built Doppler-based quality estimation for autonomous acquisition [13] and co-authored reviews on ML for robotic ultrasound and robot-assisted surgery [6, 11].
- **Open data and leadership:** served on the core team of OpenH-RF, the largest open ultrasound channel-data collection from 38 institutions [5], supervised 15+ students, and contributed to grant writing.

Visiting Researcher

03/2026 – 05/2026

Stanford University, Ultrasound Imaging and Instrumentation Lab

- Built differentiable bent-ray models for speed-of-sound imaging (UltraBend [2]).

Working Student, Mobile Software Developer

04/2021 – 12/2022

Workerbase GmbH, Munich

- Built two cross-platform industrial apps (Android, iOS, Flutter) used daily by thousands of factory workers, with clean architecture and test-driven development.

Research Assistant

09/2017 – 08/2020

Technical University of Munich

- Maintained a driving simulator for automated-driving studies and trained students.

EDUCATION

PhD in Computer Science

01/2023 – Present

Technical University of Munich · Differentiable ultrasound simulation and robotic imaging · Advisor: Prof. Nassir Navab

M.Sc. in Mechatronics and Robotics

10/2018 – 09/2022

Technical University of Munich · Focus: robot-assisted medical systems · Grade 1.5

Exchange: Universidad de los Andes, Colombia · University of Tsukuba, Japan

B.Sc. in Mechanical Engineering

10/2014 – 10/2018

Technical University of Munich · Focus: autonomous driving · Grade 2.5

Exchange: Universitat Politècnica de València, Spain

SKILLS

Programming

Python, PyTorch, JAX, C++, MATLAB, Java, Swift, Dart / Flutter

Tools

Mitsuba 3, Dr.Jit, k-Wave, Field II, ROS / Gazebo, Git, Slurm, Weights & Biases

Languages

German (native), English (fluent), Spanish (proficient), French (basic)

AWARDS & SERVICE

2026	Best Presentation Award, ASMUS @ MICCAI	2024	Best Paper Runner-Up, ASMUS @ MICCAI
2025	Reviewer Award (Hon. Mention), MICCAI	2022	Runner-Up Award, Hamlyn Winter School
2025 – 26	MCML AI-X Change Scholarship		
Reviewer	CVPR, ECCV, MICCAI, MIDL, IPCAI, IEEE T-RO, IEEE T-ASE, ASMUS, Off-Grid		

TALKS & TEACHING

Physics-Guided Ultrasound Image Formation	04/2026
Talk · Stanford University, Ultrasound Imaging and Instrumentation Lab	
(Differentiable) Ray-Based Ultrasound Simulation	07/2025
Invited talk · NVIDIA Holoscan, Advanced Workflows team	
Teaching: Computer Aided Medical Procedures I, Computational Surgineering	2023 – 2026
TUM · ultrasound imaging lectures and hands-on exercises	

PUBLICATIONS

- [1] **Felix Dülmer**, Magdalena Wysocki, Nassir Navab, Mohammad Farid Azampour. *UltraDiff: Differentiable Ray Tracing in Ultrasound for Shape Optimization*. SIGGRAPH Asia Technical Communications 2026.
- [2] **Felix Dülmer**, Samuel Beuret, Benjamin N. Frey, Mohammad Farid Azampour, Nassir Navab, Jeremy J. Dahl. *UltraBend: Refraction-Aware Propagation Modeling for Speed of Sound Imaging in Ultrasound*. ASMUS @ MICCAI 2026 · [Best Presentation Award](#)
- [3] Patris Valera, Magdalena Wysocki, **Felix Dülmer**, Mohammad Farid Azampour, Sebastian Herz, Stefan Wörz, Nassir Navab. *Multivariate Gaussian NeRF for Wide Field-of-View Ultrasound Reconstruction*. ASMUS @ MICCAI 2026.
- [4] **Felix Dülmer**, Jakob Klaushofer, Magdalena Wysocki, Nassir Navab, Mohammad Farid Azampour. *UltraG-Ray: Physics-Based Gaussian Ray Casting for Novel Ultrasound View Synthesis*. MIDL 2026.
- [5] T. S. W. Stevens*, S. W. Penninga*, W. L. van Nierop*, O. Nolan*, **Felix Dülmer**, B. N. Frey, . . . , J. Dahl, R. J. G. van Sloun, W. Simson (OpenH-RF Consortium, 181 authors). *OpenH-RF: Open Ultrasound Channel Data*. Dataset release 2026, preprint in preparation · [Core team](#)
- [6] **Felix Dülmer**, Mario Reidl, Nikolina-Alexia Fasoula, Nikoletta Katsouli, Dianye Huang, Yuan Bi, Raphael Salz, Rhiannon Lees, Zhongliang Jiang, Christoph Knappich, et al. *Translational Research and Clinical Trends in Robot-Assisted Vascular Surgery: A Narrative Overview*. Handbook of Robotic and Image-Guided Surgery, 2nd ed., Elsevier 2026.
- [7] **Felix Dülmer**, Mohammad Farid Azampour, Nassir Navab. *UltraScatter: Ray-Based Simulation of Ultrasound Scattering*. IEEE IUS 2025.
- [8] **Felix Dülmer**, Mohammad Farid Azampour, Magdalena Wysocki, Nassir Navab. *UltraRay: Introducing Full-Path Ray Tracing in Physics-Based Ultrasound Simulation*. MICCAI 2025.
- [9] Magdalena Wysocki, **Felix Dülmer**, Ananya Bal, Nassir Navab, Mohammad Farid Azampour. *UltrON: Ultrasound Occupancy Networks*. MICCAI 2025.
- [10] **Felix Dülmer**, Walter Simson, Mohammad Farid Azampour, Magdalena Wysocki, Angelos Karlas, Nassir Navab. *PHOCUS: Physics-Based Deconvolution for Ultrasound Resolution Enhancement*. ASMUS @ MICCAI 2024 · [Best Paper Runner-Up Award](#)
- [11] Yuan Bi, Zhongliang Jiang, **Felix Dülmer**, Dianye Huang, Nassir Navab. *Machine Learning in Robotic Ultrasound Imaging: Challenges and Perspectives*. Annual Review of Control, Robotics, and Autonomous Systems 2024.
- [12] Angelos Karlas, Nikolina-Alexia Fasoula, Michael Kallmayer, Christoph Schäffer, Georgios Angelis, Nikoletta Katsouli, Mario Reidl, **Felix Dülmer**, et al. *Optoacoustic Biomarkers of Lipids, Hemorrhage and Inflammation in Carotid Atherosclerosis*. Frontiers in Cardiovascular Medicine 2023.
- [13] Zhongliang Jiang*, **Felix Dülmer***, Nassir Navab. *DopUS-Net: Quality-Aware Robotic Ultrasound Imaging Based on Doppler Signal*. IEEE Transactions on Automation Science and Engineering 2023.

* shared first authorship